

Why does bagging reduce variance

And why do Random Forests do better than bagging

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Why Does Bagging Work?

Motivation

- Decision trees are typically **high-variance** estimators.
- Small changes in the training data can lead to very different trees.
- The idea of *bagging* is to:
 1. Generate many bootstrap samples.
 2. Train one tree on each bootstrap sample.
 3. Aggregate the predictions.

- But why should averaging many unstable trees improve prediction?

Bagging as an Averaging Procedure

Suppose we build B trees:

$$\hat{f}_1(x), \hat{f}_2(x), \dots, \hat{f}_B(x)$$

and define the bagged predictor as:

$$\hat{f}_{\text{bag}}(x) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(x)$$

Assume:

$$\text{Var}(\hat{f}_b(x)) = \sigma^2$$

and that all pairs of predictors have correlation:

$$\text{Corr}(\hat{f}_b(x), \hat{f}_{b'}(x)) = \rho$$

Therefore:

$$\text{Cov}(\hat{f}_b(x), \hat{f}_{b'}(x)) = \rho\sigma^2$$

Variance of the Bagged Predictor

The variance of the aggregated predictor is:

$$\text{Var} \left(\frac{1}{B} \sum_{b=1}^B \hat{f}_b(x) \right)$$

Using the variance formula for sums:

$$\text{Var} \left(\sum_{b=1}^B \hat{f}_b(x) \right) = \sum_{b=1}^B \text{Var}(\hat{f}_b(x)) + \sum_{b \neq b'} \text{Cov}(\hat{f}_b(x), \hat{f}_{b'}(x))$$

Substituting the assumptions:

$$= B\sigma^2 + B(B-1)\rho\sigma^2$$

Thus:

$$\text{Var}(\hat{f}_{\text{bag}}(x)) = \frac{1}{B^2} [B\sigma^2 + B(B-1)\rho\sigma^2]$$

Simplifying:

$$\boxed{\text{Var}(\hat{f}_{\text{bag}}(x)) = \sigma^2 \left(\rho + \frac{1-\rho}{B} \right)}$$

Interpretation of the Result

If predictors were independent

If:

$$\rho = 0$$

then:

$$\text{Var}(\hat{f}_{\text{bag}}(x)) = \frac{\sigma^2}{B}$$

- Averaging strongly reduces variance.
- Increasing B makes predictions more stable.

In practice

- Trees are **not independent**.
- Bootstrap samples overlap.
- Similar observations often generate similar trees.

Therefore:

$$\rho > 0$$

and:

$$\text{Var}(\hat{f}_{\text{bag}}(x)) = \sigma^2 \left(\rho + \frac{1-\rho}{B} \right)$$

Key Consequence

As:

$$B \rightarrow \infty$$

we obtain:

$$\text{Var}(\hat{f}_{\text{bag}}(x)) \rightarrow \rho\sigma^2$$

Therefore:

- Bagging reduces variance.
- But the reduction is limited by the correlation between trees.

The more correlated the trees are, the less variance reduction we obtain.

From Bagging to Random Forests

The Limitation of Bagging

Bagging improves prediction accuracy because:

- it averages many unstable trees,
- and averaging reduces variance.

However:

- the trees are often highly correlated,
- especially when a few predictors are very strong.

In that situation:

- the same variables tend to appear in the first splits,
- many trees become structurally similar,
- and averaging becomes less effective.

The Role of Correlation

Recall:

$$\text{Var}(\hat{f}_{\text{bag}}(x)) = \sigma^2 \left(\rho + \frac{1-\rho}{B} \right)$$

Once B is large:

$$\text{Var}(\hat{f}_{\text{bag}}(x)) \approx \rho \sigma^2$$

This shows that:

- increasing the number of trees eventually has limited impact,
- unless we also reduce the correlation ρ .

The Key Idea Behind Random Forests

Random Forests introduce an additional source of randomness.

Instead of considering all predictors at every split:

- only a random subset of predictors is considered.

This forces different trees to:

- explore different splitting variables,
- become less similar,
- and therefore become less correlated.

Why Random Forests Improve Over Bagging

Reducing tree correlation decreases:

$$\rho$$

and therefore decreases:

$$\text{Var}(\hat{f}_{\text{RF}}(x))$$

Thus Random Forests aim to:

1. Keep the low bias of deep trees.
2. Preserve the variance reduction of bagging.
3. Further improve performance by reducing correlation between trees.

Summary

Bagging

- Reduces variance by averaging many trees.
- Works especially well with unstable models.
- Performance is limited by correlation between trees.

Random Forests

- Add randomness in predictor selection.
- Reduce correlation between trees.
- Achieve additional variance reduction.
- Usually outperform pure bagging.